Example 1

```
import java.util.HashMap;
import java.util.Map;

public class A_HashMap {
   public static void main(String[] args) {
      // Create HashMap
      HashMap<Integer, String> map = new HashMap<>();

      // Add elements (key, value)
      map.put(3, "Banana");
      map.put(1, "Apple"); //
      map.put(4, "Mango");
      map.put(2, "Orange");
      map.put(1, "Orange"); //duplicate key
      // Print all entries
      System.out.println("HashMap Elements:");


      for (Map.Entry<Integer, String> entry : map.entrySet()) {
         System.out.println("Key: " + entry.getKey() + " Value: " +
entry.getValue());
      }

      // Access value by key
      System.out.println("Get value with key 2: " + map.get(2));

      // Remove by key
      map.remove(3);
      System.out.println("After removing key 3: " + map);
   }
}

//Output (order is random because HashMap does not maintain order):
```

Example 2

```
import java.util.Map;
import java.util.TreeMap;

public class B_TreeMap {
   public static void main(String[] args) {
      // Create TreeMap
      TreeMap<Integer, String> map = new TreeMap<>();

      // Add elements (key, value)
      map.put(3, "Banana");
      map.put(1, "Apple");
      map.put(4, "Mango");
      map.put(2, "Orange");

      // Print all entries
      System.out.println("TreeMap Elements:");
      for (Map.Entry<Integer, String> entry : map.entrySet()) {
         System.out.println("Key: " + entry.getKey() + " | Value: " +
entry.getValue());
      }

      // Access value by key
      System.out.println("Get value with key 2: " + map.get(2));

      // First and last key
      System.out.println("First Key: " + map.firstKey());
      System.out.println("Last Key: " + map.lastKey());
   }
}
```

Example 3

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
```

```
class Employee {
    int empNo;
    String name;
    int age;

    Employee(int empNo, String name, int age) {
        this.empNo = empNo;
        this.name = name;
        this.age = age;
    }

    @Override
    public String toString() {
        return "Employee No: " + empNo + ", Name: " + name + ", Age: " + age;
    }
}

public class C_HashMapIntegerKey {
    public static void main(String[] args) {
        // Create HashMap (Key: Employee Number, Value: Employee Object)
        HashMap<Integer, Employee> employees = new HashMap<>();

        // Add employees
        employees.put(101, new Employee(101, "Alice", 25));
        employees.put(103, new Employee(103, "Bob", 30));
        employees.put(102, new Employee(102, "Charlie", 28));
        employees.put(102, new Employee(102, "Charlie2", 28));

        // Display all employees
        System.out.println("HashMap Employees:");
        for (Map.Entry<Integer, Employee> entry : employees.entrySet()) {
            System.out.println(entry.getValue());
        }

        /*
        // Get employee by empNo
        System.out.println("\nGet Employee with empNo 102:");
```

```
        System.out.println(employees.get(102));

        // Remove employee
        employees.remove(103);
        System.out.println("\nAfter removing empNo 103: " + employees);


        HashMap<Integer, Employee> employeesResut = new HashMap<>();
*/

    }
}
```

Example 4

```
import java.util.HashMap;
import java.util.Map;
import java.util.Objects;

class Emp {

    int empNo;
    String name;
    int age;

    Emp(int empNo, String name, int age) {
        this.empNo = empNo;
        this.name = name;
        this.age = age;
    }


    @Override
    public boolean equals(Object obj) {

        Emp e = (Emp) obj;

        return this.empNo == e.empNo &&
```

```
            this.name.equals(e.name) &&
            this.age == e.age;
    }


    @Override
    public int hashCode() {
        return Objects.hash(empNo, name, age);
    }

    @Override
    public String toString() {
        return "Emp No: " + empNo +
                ", Name: " + name +
                ", Age: " + age;
    }

}

public class D_HashMapEmployeeKey {

    public static void main(String[] args) {

        // Emp is the KEY
        // String is the VALUE
        HashMap<Emp, String> employees = new HashMap<>();


        //adding duplicate key - e2 - passing same reference variable
        //removes duplicate keys
        /*
        Emp e1 = new Emp(101, "Alice", 25);
        Emp e2 = new Emp(102, "Bob", 30);



        employees.put(e1, "Developer");
```

```
        employees.put(e2, "Tester");
        employees.put(e2, "Tester2");
        */


        //same data but , diff ref variable
        /*
        Emp e1 = new Emp(101, "Alice", 25);
        Emp e2 = new Emp(102, "Bob", 30);
        Emp e3 = new Emp(102, "Bob", 30);

        employees.put(e1, "Developer");
        employees.put(e2, "Tester");
        employees.put(e3, "Tester2");
        */


        for (Map.Entry<Emp, String> entry : employees.entrySet()) {
            System.out.println(entry.getKey() + " " + entry.getValue());
        }
    }
}
```